

Seurat_tutorial

Meryem Stroosma

2025-11-23

Content

- [Introduction](#)
- [Purpose](#)
- [Data](#)
- [Packages](#)
- [Loading the data](#)

Introduction

This RMarkdown contains a brief summary of the R package Seurat. Seurat is used for the analysis of single-cell RNA sequencing data, providing tools for data preprocessing, quality control, normalization, dimensionality reduction, clustering, differential expression, and cell-type annotation. Besides analyzing a dataset Seurat can also be used to create a variety of plots to visualize the data. The summary in this RMarkdown is based on the [Seurat guided clustering tutorial](#).

Purpose

The purpose of this RMarkdown is to understand the different functionalities of Seurat, and to learn how to use Seurat for analyzing the [GSE138852](#) count matrix.

Data

The data that is used in this RMarkdown can be accessed [here](#). This is a dataset of Peripheral Blood Mononuclear Cells (PBMC) freely available from 10X Genomics. This dataset is also used in the [Seurat guided clustering tutorial](#).

Packages

Prior to analysis several packages need to be loaded. The required packages are the following:

- **dplyr**: this package is used by Seurat for data manipulation.
- **Seurat**: this package needs to be loaded prior to running any Seurat specific functions.
- **patchwork**: the patchwork package can be used to combine several plots into a single figure.
- **here**: this package is used to make sure the file paths are reproducible.

Loading the dataset and creating a Seurat object

In order to perform analysis with Seurat, the data first needs to be loaded. This is done with `Read10X()` which is a Seurat function that can be used to load 10X genomics files such as matrices and .tsv files. After loading the dataset, the function `CreateSeuratObject()` is used to create a Seurat object that contains the raw data. In this object several filters can be applied or altered, which is shown in the following sections.